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B.Tech. Degree III Semester Examination November 2019

ME 1306 MACHINE DRAWING

(2012 Scheme)

(Missing dimensions if any may be suitably assumed)

Time : 4 Hours

Maximum Marks : 100

- I. (a) Make a neat sketch of conventional representation of internal thread and external thread taking standard proportions of dimensions with $d = 30$ mm. (20)
- (b) Draw a rag end type foundation bolt for 30 mm diameter and indicate standard proportions on the drawing. (10)
- OR**
- II. The isometric view of a socket and spigot joint is shown in figure 1. Draw top half sectional elevation and bottom half sectional plan. (30)
- III. Draw sectional elevation and side view of a hydraulic pipe joint suitable to 80 mm diameter pipe. Take the thickness of pipe as 25 mm and other dimensions proportionately. (30)
- OR**
- IV. The isometric view of a plumber block is shown in figure 2. Draw right half sectional elevation and plan. (30)
- V. Details of a cross head are shown in figure 3. Draw the following assembled views showing all dimensions (i) Elevation right half in section (ii) Plan. (40)
- OR**
- VI. Details of a Ramsbottom safety valve are shown in figure 4. Draw the following assembled views showing all dimensions: (40)
 - (i) Elevation right half in section
 - (ii) Plan.

(P.T.O.)

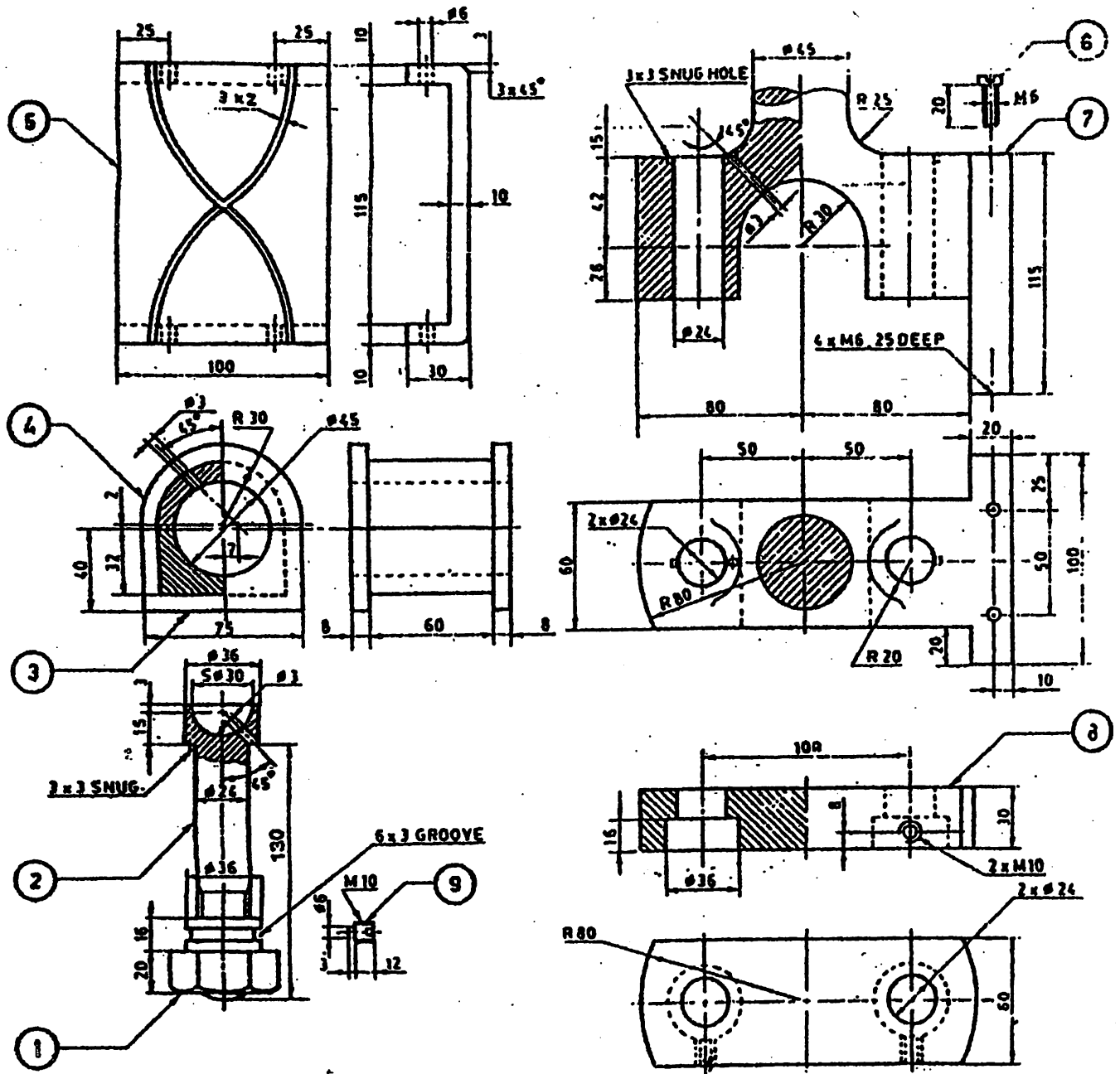


Figure 3

ITEM LIST

Item	Description	Qty	Material
1	Grooved nut	2	M.S.
2	Bolt	2	M.S.
3	Bearing (Rectangular)	1	Brass
4	Bearing (Half round)	1	Brass
5	Shoe	1	C.I.
6	Set screw	4	M.S.
7	Piston rod end	1	Steel
8	End cap	1	C.S.
9	Grub screw	2	M.S.

CROSS HEAD
(VERTICAL TYPE 2)

(P.T.O.)

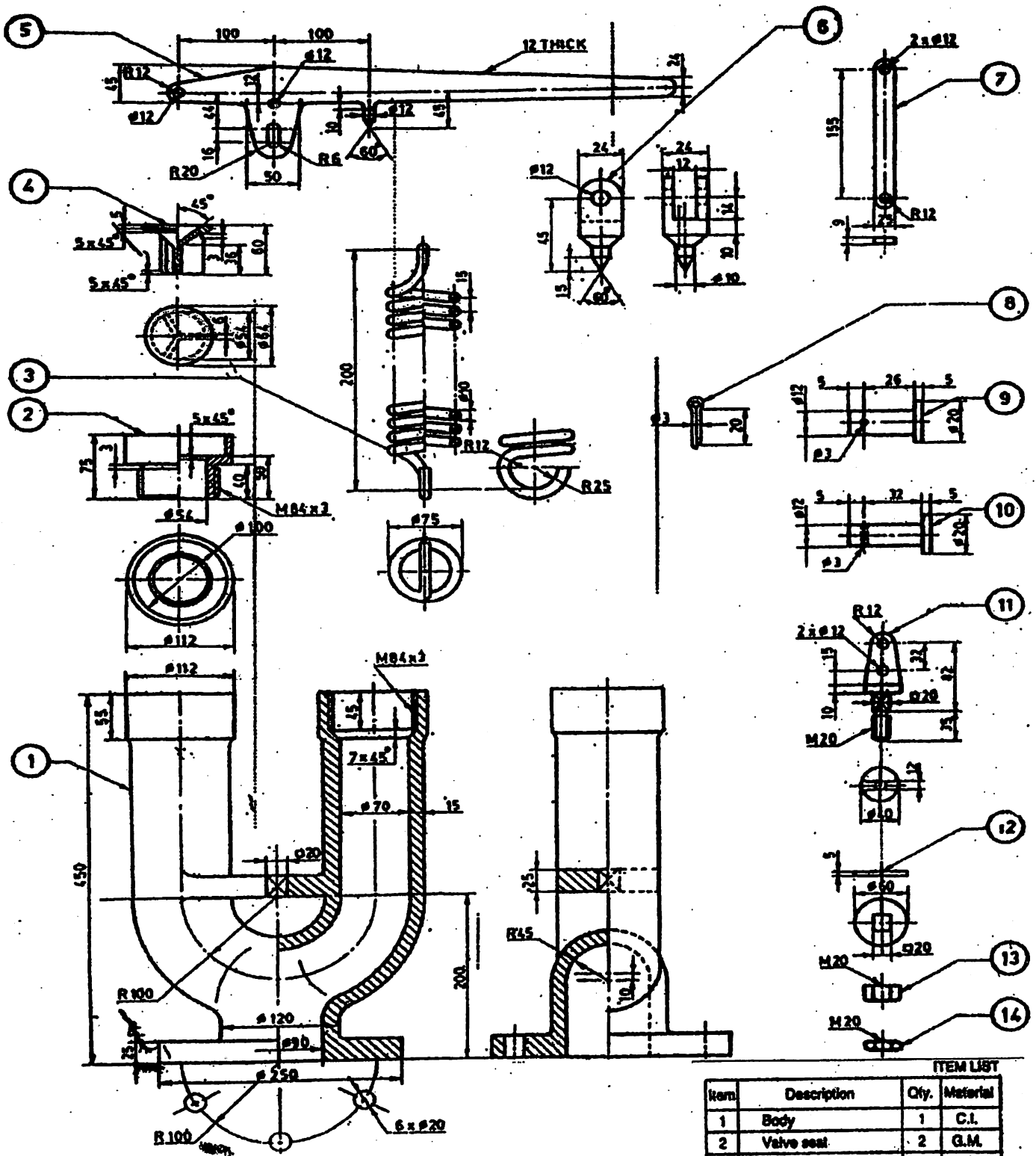


Figure 4

ITEM LIST			
Item	Description	Qty.	Material
1	Body	1	C.I.
2	Valve seal	2	G.M.
3	Spring	1	Steel
4	Valve	2	G.M.
5	Lever	1	M.S.
8	Pivot	1	M.S.
7	Link	2	M.S.
6	Split pin	3	M.S.
9	Pin for link	2	M.S.
10	Pin for pivot	1	M.S.
11	Shackle	1	M.S.
12	Washer	1	M.S.
13	Nut	1	M.S.
14	Lock nut	1	M.S.